Linear First-Order Equations & the Integrating Factor

Recognizing linear form

1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.

2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.

Finding the integrating factor

3 Find the integrating factor for $y' + 3y = e^x$.

4 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.

Solving with the integrating factor

5 Solve $y' + 3y = e^x$.

6 Solve $y' - \frac{2}{x}y = x$, $x > 0$.

Initial value problems

7 Solve $y' + y = 2$, $y(0) = 0$.

8 Solve $y' + 2y = 4x$, $y(0) = 1$.

Application: a filling tank

9 A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t, the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.